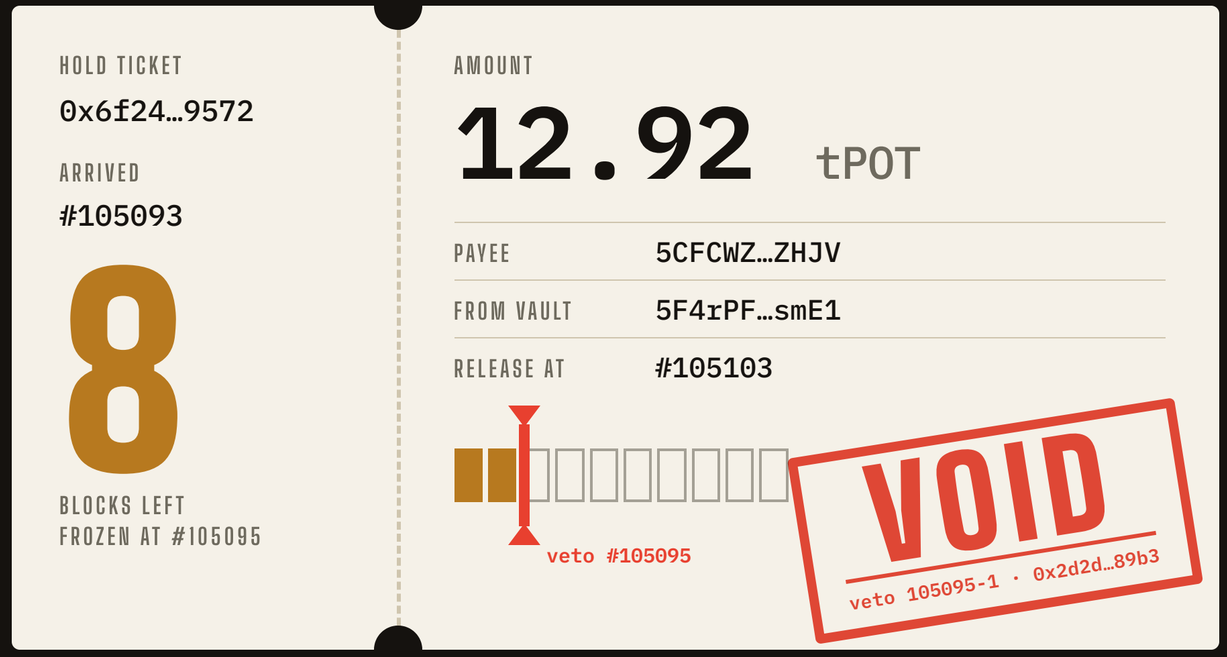
AIRLOCK

A veto window for your POT.

A stolen hot key can only announce a spend. Your cold key vetoes it before it executes.

Portaldot V3 testnet · Proxy pallet · no custom runtime code



ONE LEAKED SEED. ONE BLOCK.

POT on a single hot key leaves in one extrinsic once its seed leaks. A multisig stops that, but asks a second signer for every payment.

HOT KEY, NO DELAY

100 UNIT

moved to the thief in one extrinsic.

`control run (bench --ablate) · interim chain, not V3`

TWO-KEY MULTISIG

2 SIGNERS

on every payment, for one person.

ANNOUNCE. WAIT. VETO.

1 · ANNOUNCE → 2 · WAIT → 3 · VETO

1 · ANNOUNCE

`hot key · delay D`

It cannot spend directly. It can only publish a call hash.

```
proxy.announce(vault, hash)
```

2 · WAIT

`D = 10 blocks = 60 s on V3`

The watcher alerts: YOURS, or NOT YOURS with the amount and a veto line.

```
event Proxy.Announced
```

3 · VETO

`cold key · delay 0`

The thief's execute then fails `Proxy.Unannounced`.

```
reject_announcement(hot, hash)
```

> **Your own spend:** register the intent, announce, wait D, execute. No co-signer.

REFUSED ON V3.

HOLD TICKET
0x6f24…9572

ARRIVED
#105093

8

BLOCKS LEFT
(#105103 − #105095)

AMOUNT
12.92 tPOT

PAYEE 5CFCWZ…ZHJV

RELEASE AT #105103



VOID
veto 105095-1 · 0x2d2d…89b3

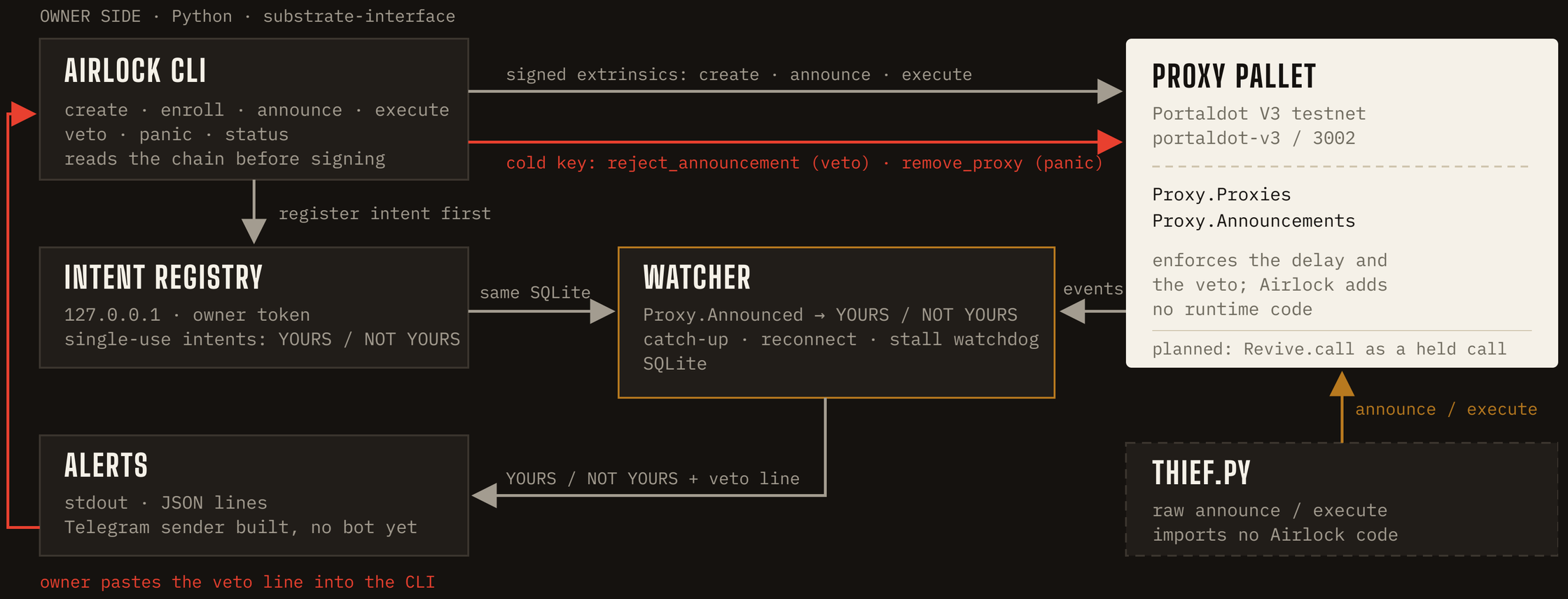
105093-1	thief Proxy.announce	ok · held	block 0xc7c3…f63b ↗
105095-1	cold reject_announcement	ok · VETO	block 0xc22c…9e26 ↗
105106-1	thief proxy_announced	Proxy.Unannounced	block 0x308b…16ea ↗
105108-1	thief proxy.proxy, direct	Proxy.Unannounced	block 0xa712…1959 ↗

excerpt · evidence/e2e-v3-2026-10-19T160547Z.log · b4d379d

```
NOT YOURS — VETO
Vault 5F4rPF…smE1
Chain: Portaldot V3 testnet (v3)
Up to 12.9200 tPOT at risk (…)
Announced by delegate 5Cwitu…ZULN at block #105093;
  executable from #105103. Veto must land by #105102:
  9 blocks (≈ 54 s) from head #105093
…
airlock veto --vault 5F4rPF…smE1 --delegate 5Cwitu…ZULN
  --hash 0x6f24…9572
…
[v3] ok  veto  Proxy.proxy(Proxy.reject_announcement)
  by cold #105095 id 105095-1 tx 0x2d2d…89b3
…
E2E PASS [('announce', 'ok'),
  ('execute', 'Proxy.Unannounced'),
  ('direct', 'Proxy.Unannounced')] · vault total
  and thief payee total unchanged: True
```

vault_total 1500000000000000 → 1500000000000000 planck
payee_total 0 → 0 planck

THE CHAIN ENFORCES. WE WATCH.



The CLI and the watcher pin the chain by genesis hash and runtime code hash before signing, and refuse an unknown chain. The thief script is deliberately independent.

THE PRIMITIVE EXISTED. THE PRODUCT DIDN'T.

Delayed proxies with announce and reject ship in the Proxy pallet. In our search (not a market survey) we found no product built on that path.

SETUP	SIGNERS PER PAYMENT	A LEAKED HOT KEY…	THE OWNER ACTS…
ONE HOT KEY	1	drains the account in one extrinsic	after the money is gone
TWO-KEY MULTISIG	2	cannot spend alone	on every payment
DELAYED PROXY, NO WATCHER	1, then wait D	can only announce; no alert goes out	only if they notice the hash on chain in time
AIRLOCK	1, then wait D	can only announce; the cold key vetoes	only on NOT YOURS

PORTALDOT'S PROXY PALLET ENFORCES IT.

`Proxy.create_pure`	the keyless vault
`Proxy.add_proxy`	hot key with delay D, cold key with 0
`Proxy.announce`	the hot key publishes a call hash
`Proxy.proxy_announced`	runs after D, if not vetoed
`Proxy.reject_announcement`	the veto
`Proxy.remove_proxy`	panic: retire the hot key

plus `Proxy.proxy` (the cold key's wrapper), `remove_announcement`, `Balances.transfer_keep_alive`

Security is Portaldot's consensus: a compromised server can mislabel or drop an alert, but cannot move funds or shorten the delay.

Running on Portaldot V3 (`portaldot-v3/3002`) since 2026-10-19. No contract deployed.

PLAN, NOT BUILT: the hold-ticket flow inside Portaldot Wallet; `Revive.call` as a held call.

WHOEVER KEEPS A HOT KEY.

Intended users. Open source today: the CLI and the watcher, MIT.

POT HOLDERS

A daily hot key for spending; the vault behind it only pays out after a veto window.

GRANTEES

Vested payouts land in a vault, so a leaked laptop key cannot sweep them.

STASHES & TREASURIES

Validator stash funds and small team wallets, one operator key each.

PLAN, NOT BUILT

FIRST USERS

grantees: new grants vested straight into the grantee's vault.

PAID

a hosted watcher with phone alerts; the CLI and watcher stay free.

WALLET

the hold-ticket flow is small enough for Portaldot Wallet to adopt.

BUILT. STILL OPEN.

ON V3 · 2026-10-19

E2E PASS	thief → NOT YOURS → veto → refused	b4d379d
7 / 7	intent registry checks, run against V3	890fcfb
25 · 8	seed extrinsics · call types	seed-v3
44 / 44	V3 proof rows re-fetched by id	PROOF.md

INTERIM CHAINS AND OFFLINE

20 / 20	bench drains vetoed, 0 planck moved	interim, not V3
350	offline tests pass · 164 commits	775c284

OPEN BEFORE THE FINAL SUBMISSION

- [ ] Benchmark on V3 (interim chains only so far)
- [ ] Phone alerts (Telegram sender built, no bot)
- [ ] Revive.call as a held call (composer only)
- [ ] Hosted owner UI (none yet; CLI today)
- [ ] Deploy the watcher + registry (local today)

CHECK IT BY EXTRINSIC ID.

SOURCE	`github.com/edycutjong/portaldot-hackathon-2026-airlock-edycu`
PROOF LOG	`docs/PROOF.md` · every row: id, block, tx
DEMO VIDEO	`youtu.be/3kbXcX7UWwU` · V3 run, ids in evidence/checkin-video-run-v3-2026-10-20.md
PAGE	`airlock.edycu.dev`
VAULT ON V3	`5F4rPF…smE1` · created 105053-1 (link opens its block)

AI assistance. Drafted with Claude (Claude Code) and reviewed by the author: most of the code, the tests, the README and this deck. The author's own: the choice of the Proxy-pallet mechanism, the vault layout and the on-chain design decisions (edycu, solo).

SCAN: SOURCE ON GITHUB